MATH120-24S2 Discrete Mathematics

Tutorial 12

References

- Topic 20, 21: Probability and Discrete Random Variables
- Sections 9.8 and 9.9 of Epp (4th or 5th Ed.)

Work on the following problems to prepare for the exam.

1. [D-B] Suppose you have a 20-sided die. You roll the die ten times. Find the probability that

   (a) the score on the first roll is divisible by 5,

   (b) the score on the first roll is divisible by 3,

   (c) the score on the first roll is divisible by 15,

   (d) the score on all ten rolls is divisible by 15, and

   (e) on at least one of the rolls, the score is not divisible by 15.

   Solution:

   The sample space for the first three parts of the question is $S = \{1, 2, \ldots, 20\}$.

   (a) Let A be the event that the score on the first roll is divisible by 5, then $A = \{5, 10, 15, 20\}$. Hence $P(A) = |A|/|S| = 4/20 = 1/5$.

   (b) Let B be the event that the score on the first roll is divisible by 3, then $B = \{3, 6, 9, 12, 15, 18\}$. Hence $P(B) = |B|/|S| = 6/20 = 3/10$.

   (c) Let C be the event that the score on the first roll is divisible by 15, then $C = \{15\}$. Hence $P(C) = |C|/|S| = 1/20$.

   Notice that A and B are not independent since $C = A \cap B$, but $P(C) \neq P(A) \cdot P(B)$.

   (d) The outcomes of each roll are clearly independent, so

   $$P(\text{all rolls are divisible by } 15) = (P(C))^{10} = \left(\frac{1}{20}\right)^{10} = \frac{1}{20^{10}}.$$

(e) The event that on at least one of the rolls, the score is not divisible by 15 is the complement of the event that the score on all ten rolls is divisible by 15. So the required probability is $1 - 1/(20^{10})$.

2. [D] Two dice are thrown. Let A be the event that the first die shows an odd score, B the event the second die shows an even score, and C the event that either both are odd or both are even. Show that

   (a) A and B are independent,

   (b) B and C are independent,

   (c) C and A are independent, but

   (d) A, B, and C are not independent.

   Solution:

   It is not absolutely necessary to define the sample space in this problem but let's do it anyway. Now

$$S = \{OO, OE, EO, EE\},$$

   where the four outcomes represent the parity of the two dice rolls in the obvious way. All four outcomes are equally likely, which is always handy.

   Clearly, $P(A) = P(\{OO, OE\}) = 1/2$, $P(B) = P(\{EE, OE\}) = 1/2$, and $P(C) = P(\{EE, OO\}) = 1/2$.

   (a) It is reasonable to claim that it is obvious that A and B are independent since they are events dealing with separate dice rolls, but let's look at it in more detail. From above, we see

$$P(A \cap B) = P(\{OE\}) = \frac{1}{4} = P(A) \cdot P(B)$$

   so A and B are independent.

   (b) $P(B \cap C) = P(\{EE\}) = \frac{1}{4} = P(B) \cdot P(C)$, so B and C are independent.

   (c) Similarly, $P(A \cap C) = P(\{OO\}) = \frac{1}{4} = P(A) \cdot P(C)$, so A and C are independent.

(d) Finally, as $A \cap B \cap C = \emptyset$,

$$P(A \cap B \cap C) = 0 \neq P(A) \cdot P(B) \cdot P(C).$$

Hence A, B, and C are not independent.

3. [B] A die is biased so that the probability that it comes up six is twice as large as each of the other probabilities.

   (a) Find the probability mass function for the score shown on the die.

   (b) Find the expected value of the score obtained after rolling the die.

   Solution:

   (a) Let X be the score obtained on a die roll. Let $c = P(X = 1)$. If $1 \leq k \leq 5$, we have $P(X = k) = c$ and $P(X = 6) = 2c$. Since $\sum_{k=1}^{6} P(X = k) = 1$, we see that $7c = 1$, so $c = 1/7$. Thus if $1 \leq k \leq 5$, then $P(X = k) = 1/7$, while $P(X = 6) = 2/7$.

   (b)

$$\begin{aligned} E(X) &= 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4) \\ &\quad + 5 \cdot P(X = 5) + 6 \cdot P(X = 6) \\ &= 1 \cdot \frac{1}{7} + 2 \cdot \frac{1}{7} + 3 \cdot \frac{1}{7} + 4 \cdot \frac{1}{7} + 5 \cdot \frac{1}{7} + 6 \cdot \frac{2}{7} \\ &= \frac{27}{7}. \end{aligned}$$

4. [B] A fair coin is flipped until either we obtain two successive heads or we have flipped the coin five times (whichever occurs sooner). Let X be equal to the number of times we flip the coin. Find the probability mass function of X and its expectation.

   Solution:

   Let X denote the number of times we flip the coin. The smallest value that X may take is two and the only sequence of tosses that gives $X = 2$ is head, head. This has probability $1/4$, so $P(X = 2) = 1/4$.

   If we toss the coin three times, then we must finish with head, head but the first toss must be tails, otherwise we would stop after two tosses. So the only sequence of tosses giving $X = 3$ is tails, heads, heads. This sequence has probability $1/8$, so $P(X = 3) = 1/8$.

If we toss the coin four times, then, as before, we must finish tails, heads, heads but this time the first toss can be whatever we like. Hence the possible sequences leading to $X = 4$ are tails, tails, heads, heads and heads, tails, heads, heads. The probability that either one or other of these sequences occurs is $1/16 + 1/16 = 1/8$. Hence $P(X = 4) = 1/8$.

Since $2 \leq X \leq 5$, we must have $P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) = 1$, so

$$\begin{aligned} P(X = 5) &= 1 - P(X = 2) - P(X = 3) - P(X = 4) \\ &= 1 - 1/4 - 1/8 - 1/8 = 1/2. \end{aligned}$$

Now

$$\begin{aligned} E(X) &= 2 \times P(X = 2) + 3 \times P(X = 3) + 4 \times P(X = 4) + 5 \times P(X = 5) \\ &= 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{8} + 5 \times \frac{1}{2} = \frac{31}{8}. \end{aligned}$$

5. [M] Suppose you are on a voyage down the River Pinn in a small boat. For provisions, you have two tins of fish, two tins of tomatoes, and two tins of custard. Unfortunately during a torrential storm, the tins are washed overboard by a large wave and, before you can retrieve them, the labels are washed off.

   (a) Show that if you open three tins, then the probability that you open one of each sort is $2/5$.

   (b) Suppose you go on opening tins until you have opened one of each sort. Find the expected number of tins that you have to open.

   Solution:

   (a) Here the sample space is all possible subsets of the six tins having size three. There are $\binom{6}{3} = 20$ of these so $|S| = 20$. Let A be the event that we open tins containing different foods. Then, as there are two possibilities for each tin, $|A| = 8$, so $P(A) = 8/20 = 2/5$.

(b) Let X denote the number of tins we have to open before we have one tin of each sort. The previous part shows that $P(X = 3) = 2/5$. First observe that we have to open at least three and at most five tins. We next find $P(X = 5)$. To do this, we will calculate the probability that the first four tins do not include all three types of food. The sample space is now all possible subsets of the six tins having size four. Thus $|S| = \binom{6}{4} = 15$. Let B be the event that the first four tins opened do not include one of each sort. Then $|B| = 3$ since the two remaining tins must be of the same type of food and there are three types. Hence $P(B) = |B|/|S| = 3/15 = 1/5$. Thus $P(X = 5) = 1/5$. Finally, as $P(X = 3) + P(X = 4) + P(X = 5) = 1$, we get $P(X = 4) = 2/5$. Hence

$$\begin{aligned} E(X) &= 3 \cdot P(X = 3) + 4 \cdot P(X = 4) + 5 \cdot P(X = 5) \\ &= 3 \times \frac{2}{5} + 4 \times \frac{2}{5} + 5 \times \frac{1}{5} = \frac{19}{5}. \end{aligned}$$

6. [B-M] Tay-Sachs disease is a rare but fatal disease of genetic origin. If a couple are both carriers of the disease, then a child of theirs has probability $1/4$ of being born with the disease independently of all others. Suppose a couple are both carriers of the disease and have five children. Find the following probabilities:

   (a) exactly one of their children is born with the disease, and

   (b) at least one of their children is born with the disease.

   Solution:

   Let X be the number of the couple's children suffering from Tay-Sachs disease. Then $X \sim \mathrm{Bin}(5, 1/4)$.

   (a)
$$P(X = 1) = \binom{5}{1}(1/4)^1(1 - 1/4)^{5-1} \approx 0.3955.$$

   (b)
$$\begin{aligned} P(X \geq 1) &= 1 - P(X < 1) = 1 - P(X = 0) \\ &= 1 - \binom{5}{0}(1/4)^0(1 - 1/4)^{5-0} \approx 0.7627. \end{aligned}$$

7. [B-M] Suppose you collect stickers of Chelsea players. In your stickers album there are 15 Chelsea players to collect.

   (a) Suppose you have so far collected 5 of the 15 stickers and that each sticker that you buy is equally likely to be of any of the 15 players. What is the probability that the next sticker that you buy is one that you don't already have? What is the distribution of the number of stickers that you must buy before you have one that is different from the ones you have so far?

   (b) Now repeat the previous part assuming that you have k stickers so far.

   (c) Let X_k be the number of stickers that you must buy in order to increase the number of different stickers in your collection from k to $k+1$. Show that $E(X_k) = 15/(15-k)$.

   (d) Hence find an expression for the expected number of stickers that you must buy in order to have a copy of each of the 15 possible stickers. Here, you may use the *linearity of expectation*, that is,

$$E\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} E(X_i).$$

Solution:

(a) There are ten stickers that you don't yet have and fifteen possible stickers so the probability that the next sticker is one that you don't have is $10/15 = 2/3$. The number of stickers that you must collect until you have one that is different from the ones you have so far has the geometric distribution with parameter $2/3$.

(b) If you have k stickers so far, then there are $15-k$ stickers still to collect, so the probability that the next sticker is one that you don't have is $(15-k)/15$. The number of stickers that you must collect until you have one that is different from the ones you have so far has the geometric distribution with parameter $(15-k)/15$.

(c) The random variable X_k has the geometric distribution with parameter $(15-k)/15$, so

$$E(X_k) = \frac{1}{(15-k)/15} = \frac{15}{15-k}.$$

(d) Let X be the number of stickers that you must buy in order to have a copy of one of each of the fifteen possible stickers. Then

$$X = X_0 + X_1 + \cdots + X_{14}.$$

So

$$\begin{aligned} E(X) &= E(X_0) + E(X_1) + \cdots + E(X_{14}) \\ &= \frac{15}{15-0} + \frac{15}{15-1} + \cdots + \frac{15}{15-14} \\ &= \sum_{k=1}^{15} \frac{15}{k} \approx 65. \end{aligned}$$